Infrastructure Management Made Easy with Salt Stack

The tremendous success of technologies like virtualisation and cloud computing has made infrastructure management a more challenging task for system administrators. This article introduces a simple and powerful tool called Salt Stack for infrastructure management.

Earlier, systems administrators had to visit each and every server to install or configure any software. This particular style of infrastructure management was very tedious and time consuming. Back then, administrators would remotely log in, a practice that was very successful for some time. Secure Shell (SSH) is a fine example for accessing a remote server. But administrators then had to establish a remote connection with every server and issue commands separately to each one. In the current cloud computing era, administrators use automation tools that can perform policy based configurations and installations on all the servers at once, automatically. For this purpose, there are a few useful open source tools like Salt Stack, Puppet, Chef, etc.

An introduction to Salt Stack

Salt (Salt-Stack) is an infrastructure automation and management system coded in Python by Thomas S. Hatch. According to official documentation, "Salt is a new approach to infrastructure management built on a dynamic communication bus. Salt can be used for data-driven orchestration, remote execution for any infrastructure, configuration management for any app stack, and much more." Hatch used the ZeroMQ messaging library to facilitate the high-speed requirements and built Salt using ZeroMQ for all networking layers. Running commands on remote systems is the core function of Salt. It can execute commands across thousands of systems, in seconds.